



34870 Lab C – Microphone calibration – E26

Room 026 – Building 354

Objectives:

- a) Measure the sensitivity of two microphones at a single frequency
- b) Measure the frequency response of two microphones, and combine with a)
- c) Fit a LTspice model to the measured responses
- d) Understand uncertainties associated with the measurement of the sensitivity

Equipment:

- Computer with Matlab
- National Instruments USB 4431 data acquisition card
- B&K microphone amplifier (Nexus)
- Actuator power supply GRAS 14AA + B&K actuator
- B&K 4133 Microphone (S/N 591628)
- B&K 4134 Microphone (S/N 1534527)
- Pistonphone (subject to availability)
- Microphone calibrator GRAS 42AG (three units)

Warnings (read before start):

- Be VERY CAREFUL when removing the microphone grid and placing the actuator on the microphone. Any contact with the diaphragm will destroy it.
- Only switch on the Actuator power supply 14AA for the measurement duration. Switch off for any manipulation.
- Do not switch on the Nexus microphone amplifier when it is connected to the microphone preamplifier. If you do so, it will read information from the preamplifier and change its settings.
- Disconnect the microphone cable from the Nexus every time you change the microphone capsule.
- LEMO cables have a sleeve with a latch. Pull it back for removal. They only fit in one position.
- The Matlab files produce figures for information, but you have to create your own figures with the measurement results.

Introduction

This exercise measures the frequency response and absolute sensitivity of a pressure-field microphone and a free-field microphone. This exercise uses a USB based data-acquisition board (National Instruments) which is operated directly from Matlab. The measurements are made using a multitone signal, which ensures a fast measurement and a good signal to noise ratio. This type of measurement is used in metrological laboratories for secondary calibration of microphones.

Evaluation of the exercise

The exercise work is done in groups of ideally three students. However, the evaluation is individual using a quiz in DTU Learn. The responses do not need to be the same for all students in the group. The quiz covers Lab B and Lab C. Submit the quiz as soon as possible and **no later than the deadline Monday 5th October.**

Part 0: Preparation

- The necessary files are available at DTU Learn and copied to the lab computer. When you complete the exercise, take the results with you and delete from the lab computer.
- Connect the output 0 of the NI card to inputs 0 and 1 using BNC cables.
- Run the script `test_meas_mag_spec2` to check the connection to the data acquisition card.

Part 1: Testing of the data acquisition system

First, the frequency response of the data acquisition system + microphone amplifier ensemble must be measured.

Connect the output channel AO0 from the NI-card to the input channel AI0 on the NI-card. Connect (using a BNC T-split) the AO0 output to the Nexus microphone amplifier input (Lemo). Use the Lemo-BNC special cable. Connect the Nexus output channel (BNC) to input AI1 on the NI-Card. See Fig. 1.

Nexus (Fig. 2):

- Use channel 1.
- Set channel 1 output value to 10mV/Pa. Set the High Pass (HP) and Low Pass (LP) filters as indicated in figure 2: 0.1 Hz and 100 kHz (lower picture).
- Set the input value (sensitivity) to 10mV/Pa. (this results in a total amplifier gain G_{Nex} of 1)
- Make sure the polarization voltage is set to 200 V.

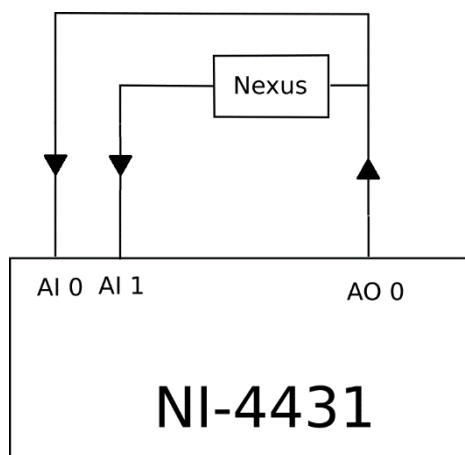


Figure 1 – Sketch for the test of the acquisition system.

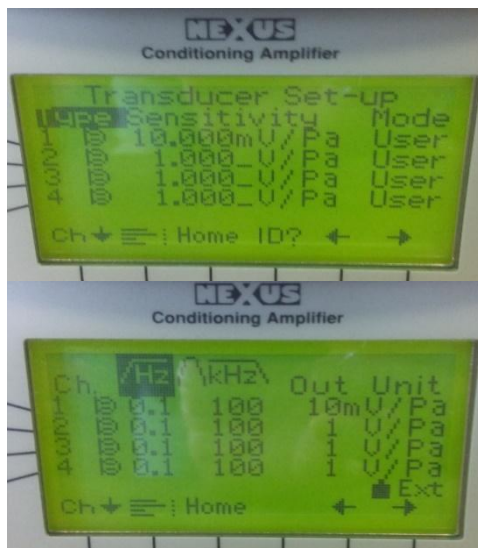


Figure 2 – Settings of the Nexus microphone amplifier.

Use the function

```
[fn,specn,f,spec]=meas_mag_spec2(f1,f2,Amax,n_oct,Nav,fb,fres)
```

```
fn:          sine frequencies
specn:       complex spectrum at sine frequencies
f:          full frequency vector
spec:       full spectrum vector
```

to measure the voltage spectrum of channels 0 and 1. Use the following parameters:

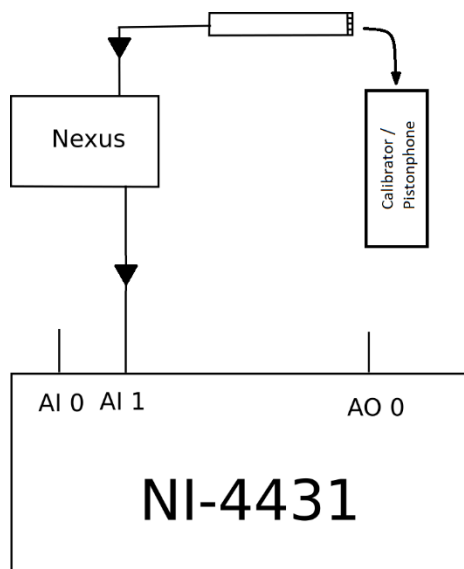
```
f1=20;          % lowest frequency
f2=60000;       % highest frequency
Amax=1;         % Maximum output signal value
n_oct=6;        % Number of signal frequencies per octave
Nav=4;          % Number of measurement averages
fb=1;           % Low frequency boost corner frequency
fres=1;         % FFT frequency resolution
```

Plot and inspect the spectrums. Calculate the transfer function

```
H_21_ref=specn(:,2)./specn(:,1)
```

between the two spectra and save it for later. We will need this reference in order to correct for the influence of the measurement system when obtaining the transfer function between two measured signals. Note down the values of H_{21_ref} at 250 Hz and 1kHz (H_{21_250} and H_{21_1000}). They will be needed in part 2.

Part 2: Calibration of the microphone sensitivity at one frequency



Connect the microphone to the Nexus amplifier. The settings of the Nexus are the same as in Part 1, but check the values again. Follow the sketch in Figure 3.

Fully insert the microphone, **slowly**, into either a calibrator or the pistonphone. Use the proper adaptors for $\frac{1}{2}$ " microphones. Leave the ensemble on the table during measurement, do not manipulate it. Switch on the calibrator/pistonphone just before starting the measurement.

Figure 3 – Sketch for the sensitivity calibration.

Use the function call

```
M=Calibrate(f,SPL,G_Nex)
%f:          signal frequency (typically 250 Hz or 1000 Hz)
%SPL:        sound pressure (typically 94 dB or 114 dB)
%G_Nex:      Nexus amplifier gain (has been set to 1)
```

to measure the microphone sensitivity for either a calibrator or the pistonphone. All the devices (calibrators and pistonphone) have a calibration document or sticker. As any other acoustic measurement device, they have been calibrated themselves. The information includes the frequency at which they operate, and the SPL they produce in their interior when a microphone is introduced. Use these values (f , SPL), which can be different for every device, as inputs in the function call above. The gain (G_{Nex}) of the Nexus amplifier should be set to 1, as a result of its settings. For example:

```
M_1kHz=Calibrate(1000,94,1)/abs(H_21_1000);
```

Calibrate both microphones with all calibration devices. Note that we use the correction obtained in Part 1 (H_{21_250} , H_{21_1000}) in order to remove the effect of the measurement system.

You will obtain several results for each microphone. Briefly analyse these results from the metrological point of view, reflecting on repeatability, reproducibility, uncertainty and true value.

Part 3: Microphone frequency response with actuator

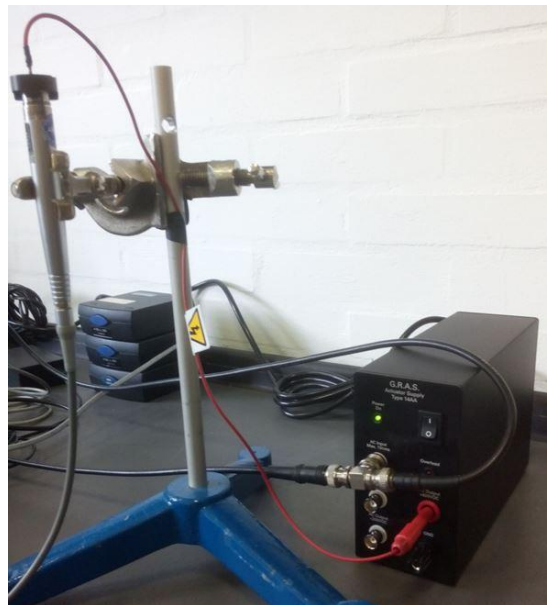
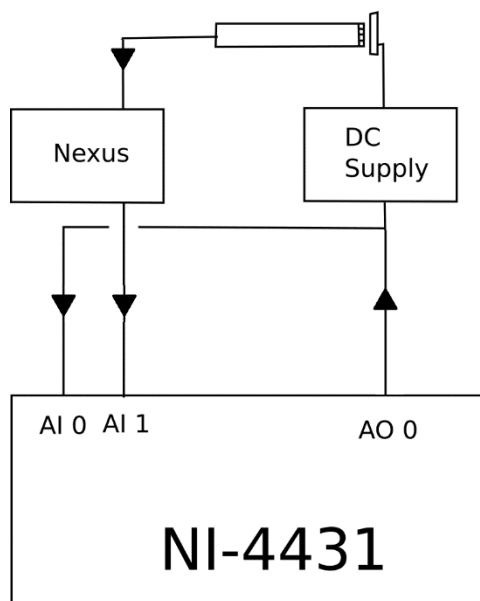


Figure 4 – Sketch and photograph of the measurement of the microphone response with an actuator.

a) Preparation of the setup and measurement issues:

- Connect the NI output AO0 to the GRAS 14AA actuator amplifier input (DC supply) (see Figure 4), and to the input channel AI0 on NI card.
- Disconnect the microphone from the Nexus (**Important!**).
- Connect the Nexus output to the input channel AI1 on the NI card.
- Make sure the GRAS 14AA actuator amplifier is turned off (**Important!**).
- Screw one of the microphone capsules to be measured to the preamplifier in the stand, as shown in Figure 4.

- The protection cap of the microphone must be unscrewed **carefully**. **Do not touch the diaphragm!**
- Connect the red wire to the actuator and **carefully** place the actuator on top of the microphone. This is best achieved by setting one leg first on the microphone rim and then slowly lowering the actuator to a horizontal position fitting the whole rim. **Careful:** the red wire might pull the actuator away when you release it!
- The other end of the red wire is connected to the GRAS 14AA actuator amplifier.
- Turn on the GRAS 14AA actuator amplifier (only for Part 3, section c). Always turn off the GRAS 14AA actuator amplifier when you are not measuring.
- Connect the microphone to the Nexus. The Nexus must be already on, otherwise it would read the microphone data and change its settings.

Note: The microphone will pick up any sound during the measurement. You must therefore **keep quiet** during the measurement.

b) Measurement of the noise floor

The noise floor is measured to ensure that the microphone responses, measured later, are well above the background level. We only need to verify that the noise level is low enough; the actual response has no other practical use. To do so, follow the guidelines in the previous section, but leave the GRAS 14AA turned off (no actuator signal). Use `meas_mag_spec2` (same procedure as in Part 1) to measure the spectrum of both the input signal of the GRAS 14AA and the microphone signal.

Use the following parameters:

```
f1=20;
f2=60000;
Amax=1;
n_oct=6;
Nav=4;
fb=200;
fres=1;
```

Save the complex spectra for later comparison with microphone measurements. Increase `Nav` to 64 and repeat the measurement, save the complex spectra as well. Remember to close the door and stay completely quiet during the measurement, as the microphone will pick up any noise in the room.

c) Measurement of the sensitivity responses of the microphones

Turn on the GRAS 14AA and repeat the measurement. Calculate the transfer function H_{21} between microphone output and excitation. Correct for the effect of the measurement system using the results from part 1:

$$H_{pv} = H_{21} / H_{21_ref}$$

Repeat the procedure for both microphones. Compare the two responses and determine which of the microphones is designed for free-field measurement and which is designed for pressure-field measurements (cavity). Re-scale the sensitivity responses to have the correct sensitivity, by using the single-frequency calibration made in Part 2 (which of the sensitivity values should you use?).

Analysis:

- Find the resonance frequency (from the phase response) and Q-factor (from the amplitude response) for both microphones. Use the procedure in the course slides (Lectures 4 & 6).
- Implement a LTspice model of the condenser microphone, using the following data:
 - Diaphragm to electrode distance: 20.77 μm . Polarization voltage: 200 V
 - Effective diaphragm diameter: 8.95 mm
 - Diaphragm mass: 1.5 mg
 - Diaphragm compliance: 0.02 mm/N,
 - Diaphragm damping: 0 Ns/m
 - Back volume: 126.4 mm³
- Use the f_s and the Q factor to find the initial values for the acoustic resistance and mass of the backplate for the two microphones
- Adjust the values in order to get the best approximation to the measured responses (magnitude and phase), if needed. Usually, no or little adjustment is needed.
- Do not use the Gpb generator implementing the T(s) function: no sound field with actuator!
- Plot the simulation and measurement in the same graph for comparison. Evaluate the agreement between model and measurement.
- Note that you may get a 180 ° difference between the phases of measurement and simulation. This is due to a different ground convention in the model and has no practical significance.

Appendix

Comments regarding the LTspice simulation of condenser microphones

When using the values given in Lab C, the output of the microphone at low frequencies is somewhat lower than the specified sensitivity of 12.5 mV/Pa. This is due to limitations in the model. According to the model in Leach the sensitivity is:

$$M = EC_{MT} \frac{S_D}{x_0} \quad \text{and the capacitance is} \quad C_{E0} = \epsilon_0 \frac{S_D}{x_0}$$

Here it is assumed that the effective area in the capacitance is the same as the effective acoustic area, which is not completely true, as the effective area of the back plate is somewhat smaller than the diaphragm area due its smaller radius and the holes.

According to the equations, two microphones with the same capacitance and sensitivity must also have the same total mechanical compliance. The two microphones (4133 and 4134) are both specified as having a sensitivity of 12.5 mV/Pa and a capacitance of 19 pF. This seems to suggest that they have the same ratio between area (S_d) and distance (x_0). However, on average, the total compliance differs by approximately 30 percent between type 4133 and 4134 and the distance differs by 10 percent. A consequence of all this is that the simple model is not able to correctly predict the sensitivity if the capacitance is locked at the nominal value of 19 pF. Thus, the values given for the simulation in Lab C are only approximations to the real values.

For the purpose of the exercise, the absolute sensitivity is less important, compared to the behavior around the resonance frequency, and therefore the lower sensitivity should just be accepted.